32-3 Bias in AI:

- Refers to systematic errors in artifical intelligence system that lead to unfair, skewed, or discriminary outcomes.

What it means for an AI Model:

- Is considered biased when it consistently produces results that favor or disadvantage certain groups of people.

Real world examples:

Hiring algorithms, eg Hiring data that would include male candidates & Not women.

Potential Consequences:

- Discrimination, Reinforcing inequality, Loss of trust

33. Key Differences

	Role	Main focus	Typical tasks	Key Skills	Common Tools
①	Data Analyst	understanding & explaining data	cleaning data	SQL, Excel	Power BI
②	Data scientist	Building predictive Models & insights	Machine learning	~~play~~ python/R	Notebook
③	Data Engineer	Building & Maintaining data infrastructure	Creating data pipelines	SQL, ETL processes	Apache Spark.
④	AI Engineer	Developing AI-Powered systems & Applications	Deploying ML models	Machine learning	Tensor-flow, API's

34. Business Intelligence (BI): is the process of collecting, analyzing & presenting data to help business make informed decisions.

Two Popular BI tools:

1. Power BI: is a business analytics tool that allows users to connect to multiple data sources.

How it helps business: Combines data from different systems e.g sales, finance

② Tableu : focuses on powerful data visualization, making complex data easy to understand through charts & interactive visuals.

How it helps businesses :

- Turns large datasets into clear, visuals insights.

e.g A marketing team can use it to analyze campaign performance.

Why BI matters :

- Make faster, evidence-based decisions.
- Identify opportunities & risks early

35. AI cause it inherits and amplify human biases that exist in the data it is trained on.

36. Different roles eg Data Scientist, Data Engineer, their responsibilities overlap in real companies.

Resources used to help → Introductory articles & career guides, youtube.

37. They are shaping how work gets done, & will have impact on both my future career & everyday life.

E.g I could use AI tools to quickly clean datasets, detect patterns.

Overall impact : The biggest shift is that AI will not just be a tool I use occasionally.

References :

① Author : Misha SV

Organization : Towards data science

Year : 24/12/2022

Web : https://towardsdatascience.com

② Author : FORTINET

Organization : Fortinet cve analysis

Year : 2026/01/13

③ Author : Jiya Lalaiya

Organization : Ira skills

Year : 2025/03/28

weblink : https://iraskillsrai

④ Author: Nick Zhao

Organisation: Vinchin

Year: 2025/09/22

Weblink: https://www.vinchin.com